

CSA17 – Artificial Intelligence: Assessment Tool 4 (Solutions)

COURSE: Artificial Intelligence (CSA17) — Assessment Tool 4 CO3: Knowledge Representation and Reasoning — Resolution Algorithm Implementation Total Marks: 30 (5 Questions × 6 marks each)

Note on mark split: The paper gives 5 questions for 30 total marks (**6 marks per question**) but does not print a value against each task. Each question sets 5 tasks in the same pattern (represent in PL, convert to CNF, negate the goal, apply resolution, state the empty clause/conclusion). Based on the rubric weighting (Problem Understanding 20%, Method Selection 20%, Solution Process 25%, Accuracy 20%, Interpretation 15%), the 6 marks per question are split below as: **(a) PL representation — 1, (b) CNF conversion — 1, (c) Negate the goal — 1, (d) Resolution steps — 2, (e) Empty clause & conclusion — 1.** Adjust if your instructor specifies a different breakdown.

Question 1 — Rain and Wet Ground (6 marks)

KB: (1) If it rains, the ground becomes wet. (2) It is raining. **Goal:** Prove the ground is wet.

(a) Propositional Logic Representation — 1 mark

Let R = “It rains”, W = “The ground is wet”

- Rule: $R \rightarrow W$
- Fact: R

(b) Convert to CNF — 1 mark

- $R \rightarrow W = \neg R \vee W$ (already CNF)
- R (unit clause, already CNF)

(c) Negate the Goal — 1 mark

Negated goal: $\neg W$

(d) Apply Resolution — 2 marks

Clause set: $C1 = \neg R \vee W$, $C2 = R$, $C3 = \neg W$ (negated goal)

1. Resolve $C1 (\neg R \vee W)$ with $C2 (R)$ $\rightarrow W$
2. Resolve W (step 1) with $C3 (\neg W)$ $\rightarrow \square$ (empty clause)

(e) Empty Clause & Conclusion — 1 mark

The empty clause $\square$ is derived, so $KB \wedge \neg \text{Goal}$ is unsatisfiable. **Conclusion: “The ground is wet” is PROVED.**

Question 2 — Student Assignment Submission (6 marks)

KB: (1) If a student submits the assignment, the student receives internal marks. (2) Rahul submitted the assignment. **Goal:** Prove Rahul receives internal marks.

(a) Propositional Logic Representation — 1 mark

Let S = “Student submits the assignment”, M = “Student receives internal marks”

- Rule: $S \rightarrow M$
- Fact: S (specifically, for Rahul: Submitted(Rahul))

(b) Convert to CNF — 1 mark

- $S \rightarrow M = \neg S \vee M$
- S (unit clause)

(c) Negate the Goal — 1 mark

Negated goal: $\neg M$

(d) Apply Resolution — 2 marks

Clause set: $C1 = \neg S \vee M$, $C2 = S$, $C3 = \neg M$

1. Resolve $C1$ ($\neg S \vee M$) with $C2$ (S) $\rightarrow M$
2. Resolve M (step 1) with $C3$ ($\neg M$) $\rightarrow \square$ (empty clause)

(e) Empty Clause & Conclusion — 1 mark

Empty clause derived $\rightarrow$ contradiction with $KB \wedge \neg \text{Goal}$. **Conclusion: “Rahul receives internal marks” is PROVED.**

Question 3 — Library Membership (6 marks)

KB: (1) If a person is a library member, the person can borrow books. (2) Priya is a library member. **Goal:** Prove Priya can borrow books.

(a) Propositional Logic Representation — 1 mark

Let L = “Person is a library member”, B = “Person can borrow books”

- Rule: $L \rightarrow B$

- Fact: L (for Priya: LibraryMember(Priya))

(b) Convert to CNF — 1 mark

- $L \rightarrow B = \neg L \vee B$
- L (unit clause)

(c) Negate the Goal — 1 mark

Negated goal: $\neg B$

(d) Apply Resolution (until no new clauses can be generated) — 2 marks

Clause set: $C1 = \neg L \vee B$, $C2 = L$, $C3 = \neg B$

1. Resolve $C1 (\neg L \vee B)$ with $C2 (L)$ $\rightarrow B$
2. Resolve B (step 1) with $C3 (\neg B)$ $\rightarrow \square$ (empty clause)

No further clauses can be generated after $\square$ is reached — resolution terminates.

(e) Final Conclusion — 1 mark

The empty clause $\square$ confirms the negated goal is inconsistent with the KB. **Conclusion: “Priya can borrow books” is PROVED.**

Question 4 — Placement Eligibility (6 marks)

KB: (1) If a student clears the aptitude test, the student is eligible for placement. (2) Arun cleared the aptitude test. **Goal:** Prove Arun is eligible for placement.

(a) Propositional Logic Representation — 1 mark

Let C = “Student clears the aptitude test”, E = “Student is eligible for placement”

- Rule: $C \rightarrow E$
- Fact: C (for Arun: Cleared(Arun))

(b) CNF Clauses — 1 mark

- $C \rightarrow E = \neg C \vee E$
- C (unit clause)

(c) Negate the Goal — 1 mark

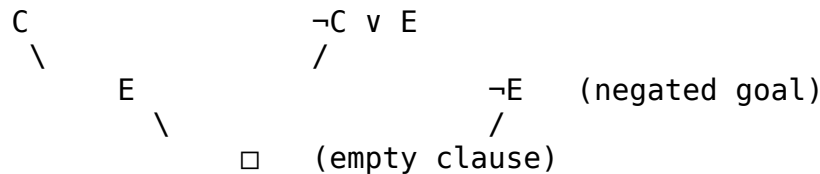
Negated goal: $\neg E$

(d) Apply Resolution — 2 marks

Clause set: $C1 = \neg C \vee E$, $C2 = C$, $C3 = \neg E$

1. Resolve C1 ($\neg C \vee E$) with C2 (C) $\rightarrow E$
2. Resolve E (step 1) with C3 ($\neg E$) $\rightarrow \square$ (empty clause)

(e) Resolution Tree & Conclusion — 1 mark



Conclusion: “Arun is eligible for placement” is PROVED.

Question 5 — Access Control System (6 marks)

KB: (1) If a user enters the correct password, the user is authenticated. (2) If a user is authenticated, the user is granted access. (3) The user entered the correct password. **Goal:** Prove the user is granted access.

(a) Propositional Logic Representation — 1 mark

Let P = “User enters the correct password”, A = “User is authenticated”, G = “User is granted access”

- Rule 1: $P \rightarrow A$
- Rule 2: $A \rightarrow G$
- Fact: P

(b) Convert Implications to CNF — 1 mark

- $P \rightarrow A = \neg P \vee A$
- $A \rightarrow G = \neg A \vee G$
- P (unit clause)

(c) Negate the Goal — 1 mark

Negated goal: $\neg G$

(d) Apply Resolution to Derive the Empty Clause — 2 marks

Clause set: C1 = $\neg P \vee A$, C2 = $\neg A \vee G$, C3 = P, C4 = $\neg G$

1. Resolve C1 ($\neg P \vee A$) with C3 (P) $\rightarrow A$
2. Resolve A (step 1) with C2 ($\neg A \vee G$) $\rightarrow G$
3. Resolve G (step 2) with C4 ($\neg G$) $\rightarrow \square$ (empty clause)

(e) Step Explanation & Final Conclusion — 1 mark

- Step 1: Since the user entered the correct password (P is true) and “password correct implies authenticated” ($\neg P \vee A$), we resolve to derive **A** (the user is authenticated).
- Step 2: Since the user is authenticated (A is true) and “authenticated implies granted access” ($\neg A \vee G$), we resolve to derive **G** (the user is granted access).
- Step 3: G contradicts the negated goal $\neg G$, producing the empty clause $\square$ — confirming $KB \wedge \neg \text{Goal}$ is unsatisfiable.

Conclusion: “The user is granted access” is PROVED.

Summary of Marks

Question	(a) PL Rep.	(b) CNF	(c) Negate Goal	(d) Resolution	(e) Conclusion	Total
1. Rain and Wet Ground	1	1	1	2	1	6
2. Assignment Submission	1	1	1	2	1	6
3. Library Membership	1	1	1	2	1	6
4. Placement Eligibility	1	1	1	2	1	6
5. Access Control System	1	1	1	2	1	6
Grand Total						30